Beijing University of Chemical Technology

2021-2022 2nd term Final Exam paper (A)

for Chemical Product Engineering

Course ID: CHE 41200T

Class:_____ Student name:_____ Student No:_____

Teacher Name: _____

Question NO	1	2	3	4	Total score
Score					

NOTE: Please answer all questions ***in English***, and finish in **120 min**.

1. Please judge the following statements, or choose appropriate answer.（30 pts）

1) (　　) Chemical product engineering is a systematic procedure or framework aimed to design new products with specified functionality to meet the market demands as soon as possible rather than to make the product with least cost.

(a) True　(b) False　(c) Uncertain

2) (　　) Product design is generally composed of four steps: (1) needs identification; (2) ideas generation; (3) ideas selection; and (4) _____

(a) process design (b) product manufacture (c) patent application (d) risk analysis

3) (　　) Which of the following does not belong to a chemical product as defined in Chemical Product Engineering?

（a） PVC plastics (b)additives of plastics (c)perfume (d)chemical simulation software

4) (　　) In addition to the real products, chemical products may be also a virtue one, for example.

(a) *Chemical simulation software (b) virtual reality glasses (c) Augmented reality glasses*

5) (　　) Thermodynamics is a useful tool to judge the possibility of a newly designed chemical product.

(a) True (b) False (c) Uncertain

6) ( ) For a new chemical product, the production process efficiency and energy integration may be less important than its speed to reach the marketplace.

(a) True (b) False (c) Uncertain

7) **( )** In order to cut down the number of raw ideas from 20 to 5, a semi-quantitative method, screening matrix, is often used.

(a) True (b) False (c) Uncertain

8) ( ) The problem solving style may be adaptive or innovative, which depends on discipline, personnel attributes as well as the problems to be solved. The development process of the penicillin series drugs from penicillin is innovative.

(a) True (b) False (c) Uncertain

9) ( ) In the product selection process, the risk level should be assessed based on the product of risk possibility and risk consequence.

(a) True (b) False (c) Uncertain

10) ( ) The economy of a new chemical product is often evaluated by the net present values (NPV), which means the product's total value in today's dollars, including all future cash flows.

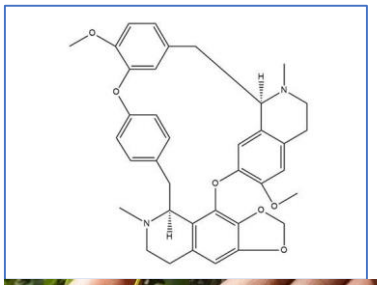
(a) True (b) False (c) Uncertain

2. **Please** describe the general procedure of a chemical product design, and illustrate briefly the main objectives of each step. (**20 points**)

3. What are the micro-structured chemical products? What are their features or characteristics? Please illustrate them with one example. (2**0 points**)

4. **How to developed a new chemical drug for COVID-19?** (**30 points**)

COVID-19 is a frustrating epidemic, which has lasted for about three years worldwide and led over millions of death tolls. The global scientists have been working hard to find an effective medicine to cure or prevent its spreading. Recently, professor Tong Yigang, from college of life science and technology of Beijing University of Chemical Technology, has disclosed a breaking news. He found that the extract(提取物) of Qian-Jin-Teng-Su (a Chinese herb medicine, see the right picture), can efficiently prohibit the replication of COVID-19 virus and has applied a patent. Its 15390-





fold(倍) efficacy is outstanding and superior to most of the available medicines like *Remdesivir* and *PAXLOVID* in the market. Now, a drug company wants to produce this medicine, and you are supposedly to be assigned as the head of the product development team.

1) Please list all the work to be conducted before production.

2) What kinds of missing information should be supplied further?

3) What other issues should be considered for its commercial production?